

Computação e Multimídia, Project Phase 1

Specification & State of the Art

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1 The Magic Cube

We propose **The Magic Cube** – a media that reimagines how we explore and experience visual content. Magic Cube is a three-dimensional artifact where each of its six faces reveals your photos and videos organized through a different perception.

1.1 Core Concept: Six Faces, Six Realities

The Magic Cube transforms based on how you look at it. Each different combination of cube rotation represents a different perceptual organization of the entire media collection:

- **Light Face** – Organized by mean luminance (darkest to brightest). Images cascade from shadow to sunshine.
- **Contrast Face** – Sorted by luminance variance (soft gradients to sharp contrasts). Quiet to dramatic.
- **Hue Face** – Sorted by hue variance (groups pictures with major in one color/set of colors).
- **Detail Face** – Arranged by edge density (smooth surfaces to intricate details). Simple to complex.
- **Texture Face** – Grouped by texture similarity. Glass-like smoothness to rough grain.
- **Energy Face** – Ordered by motion energy for videos (still meditation to kinetic chaos).
- **Kinship Face** – Clustered by visual similarity using ORB keypoints. Finds photos of the same places/subjects.

The Magic: Rotate the cube with a swipe gesture, or through the keyboard, and watch your entire media library reorganize itself in real-time. Per example: a sunset photo - On the Light face it's near the top; on the Kinship face it's clustered with other vacation shots; on the Contrast face it's surrounded by other dramatic silhouettes. The cube doesn't just organize, it reveals hidden patterns in your pictures.

We also have these actions:

- High-motion videos pulse and ripple, sending energy waves across the cube face, or making it tremble
- The cube rotates and feels weighted, physical, like a real object

2 Specification

2.1 Visual Feature Extraction

Every image and video is analyzed to extract perceptual features that drive spatial behavior and organization. All metadata is stored in XML format for consistency across the system.

Core Features:

- **Mean Luminance** – Drives Light Face organization; controls glow intensity; bright images float higher

- **Luminance Variance** – Drives Contrast Face; high variance media appears larger and more prominent
- **Edge Density / Edge Histograms** – Organizes Detail Face; influences clustering distance—detailed images group tighter
- **Texture Features** – Powers Texture Face; enables similarity grouping (smooth vs. rough surfaces attract)
- **ORB Keypoints** – Drives Kinship Face; enables visual matching—finds images of same subjects/scenes
- **Motion Energy (Video)** – Organizes Energy Face; controls pulse speed—fast videos ripple, slow ones drift
- **Video Rhythm** – Synchronizes with spatial audio; creates rhythmic patterns during playback

2.2 Gesture-Based Cube Manipulation

The cube responds to intuitive, touchless gestures detected through webcam using face detection, motion detection, or simple gesture recognition (or through keyboard input WASD):

- **Swipe Left/Right** – Rotate the cube horizontally to reveal different faces and reorganization schemes
- **Swipe Up/Down** – Rotate the cube vertically, exploring opposite perspectives
- **Strong Shake Motion** – Trigger dramatic reorganization—watch media fly to new positions based on current face's rule
- **Forward Reach** – Dive inside the cube for different combinations from a different perspective, being in a new world
- **Face Tracking** – The cube subtly tilts to follow your position, enhancing depth perception



Figure 1: Fig.1 Prototype of project with motion panel and effect examples

2.3 Immersion & Presence Techniques

To create the sensation of being physically present inside the cube:

- Camera positioned at human eye height within the cube’s interior space
- Dynamic spotlights illuminate the currently focused face, dim the others
- Floating metadata labels hover near images, tethered like magical tags
- Spatial audio—videos sound louder when their face rotates toward you (audio intensity based on distance)
- Particle effects during transitions—images leave light trails as they reorganize

2.4 User Interface Elements

Users should feel immersed in media, not navigating UI chrome:

- **Face Indicator (top center)** – Shows current active face (Light / Contrast / Detail / Texture / Energy / Kinship)
- **Gesture Feedback (bottom center)** – Subtle icon showing detected gesture state
- **Floating Metadata** – Appear near focused frames (filename or dominant feature values)
- **Connection Threads** – Visible during reorganization, showing similarity relationships between media

2.5 Interaction Flow

1. User enters experience, cube floats in center, Light face visible, media organized by brightness
2. Swipe gestures rotate cube smoothly so new faces come into view
3. On each face, media reorganizes itself according to that face’s perceptual principle
4. Shake gesture triggers dramatic reorganization, connection threads appear, media animates to new positions
5. Forward gesture zooms into selected frame for fullscreen viewing, from inside the cube, with play/pause controls
6. User explores different faces, discovering hidden patterns and visual connections

3 Technical Architecture

3.1 System Components

The system comprises four core subsystems:

- **Feature Extractor** – Analyzes images/videos, computes all 7 visual features, stores metadata in XML format
- **Cube Manager** – Manages cube geometry, face layouts, rotation state, and reorganization animations
- **Gesture Detector** – Processes webcam input, detects swipes/shakes/reaches, translates to cube commands
- **Immersion Renderer** – Handles fog, dynamic lighting, glow effects, particle trails, and spatial audio

3.2 Preliminary Class Structure

- **MagicCube** – Main controller managing cube state, rotation, and current active face
- **CubeFace** – Represents one face with its organization rule and media layout
- **MediaFrame** – Base class for image/video positioned in 3D space on a cube face
- **FeatureVector** – Encapsulates all 7 computed visual features for one media item
- **GestureDetector** – Processes webcam frames for swipe, shake, reach, and face tracking
- **CameraController** – Manages virtual camera with smooth inertial movement and face tracking
- **SimilarityEngine** – Computes distance metrics between feature vectors for Kinship face
- **XMLMetadataStore** – Reads/writes feature data to XML files, validates schema
- **EffectsRenderer** – Handles fog, lighting, glow, particles, and audio spatialization

3.3 Distance Function for Similarity

The system implements a distance function that computes similarity between media items based on their feature vectors. This enables:

- **Automatic grouping** of visually similar media on the Kinship face
- **Spatial reorganization** based on similarity when switching between faces
- **Navigation toward similar items** – moving to one image suggests related content nearby

The distance metric combines weighted contributions from texture features and ORB keypoint matches to identify media depicting the same subjects, scenes, or visual characteristics.

3.4 Technical Foundation

The implementation will build upon established techniques:

- OpenCV for feature extraction (luminance, variance, edge detection, ORB keypoints)
- Processing/OpenFrameworks for 3D rendering and easyCam integration
- Computer vision libraries for webcam-based gesture recognition
- XML for structured metadata storage ensuring consistency between analysis and display